Basics of Chemical Machine Learning, AQME and Descriptor Generation

The AI “hype”

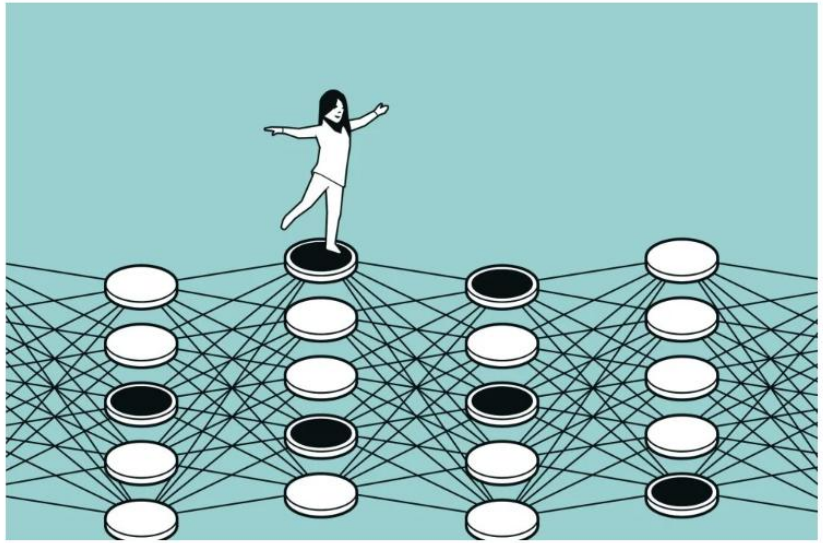
The Nobel Prize in Physics 2024

They used physics to find patterns in information

This year’s laureates used tools from physics to construct methods that helped lay the foundation for today’s powerful machine learning. John Hopfield created a structure that can store and reconstruct information. Geoffrey Hinton invented a method that can independently discover properties in data and which has become important for the large artificial neural networks now in use.

Related articles

Press release
Popular information: They used physics to find patterns in



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The AI “hype”

The Nobel Prize in Chemistry 2024

They cracked the code for proteins’ amazing structures

The Nobel Prize in Chemistry 2024 is about proteins, life’s ingenious chemical tools. David Baker has succeeded with the almost impossible feat of building entirely new kinds of proteins. Demis Hassabis and John Jumper have developed an AI model to solve a 50-year-old problem: predicting proteins’ complex structures. These discoveries hold enormous potential.

Related articles

Press release
Popular information: They have revealed proteins’ secrets through

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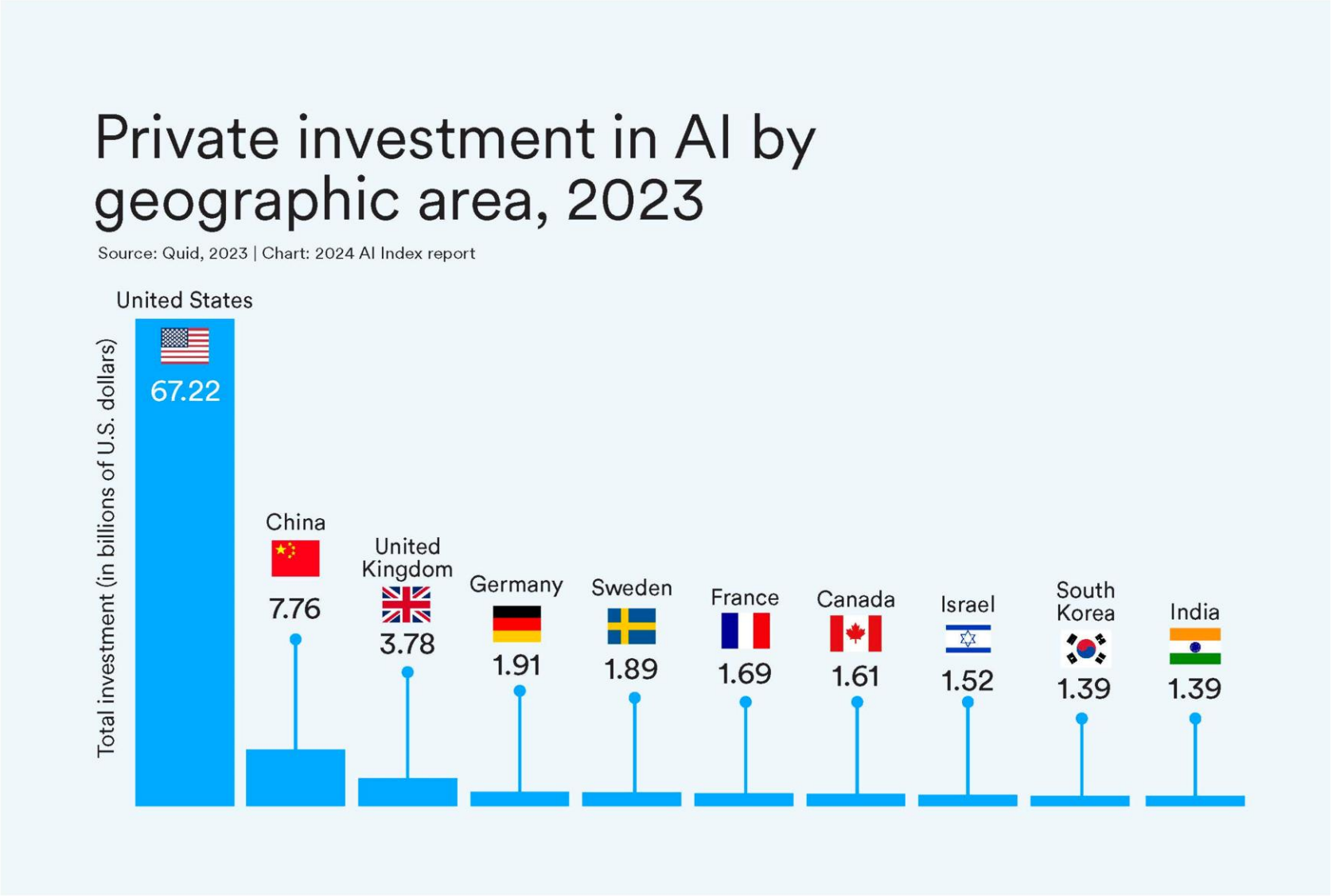
The AI “hype”



Stanford AI Index Report 2024

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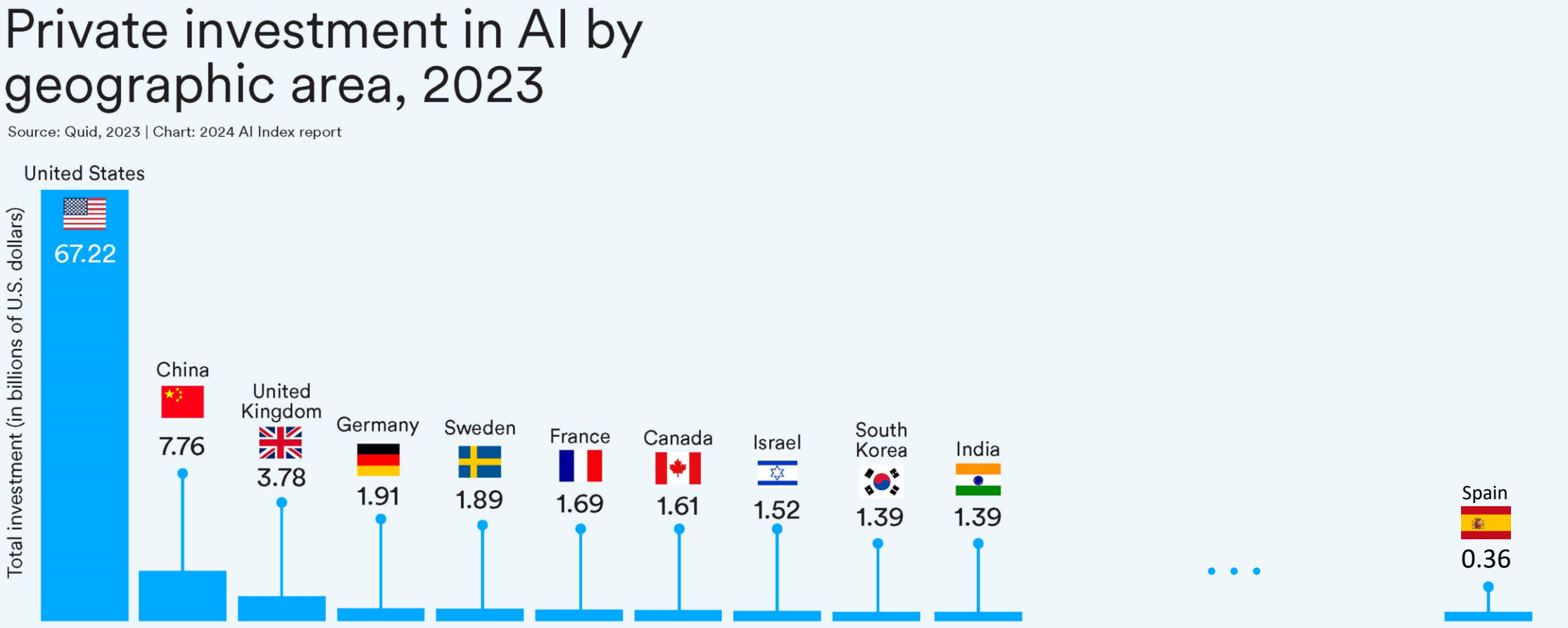

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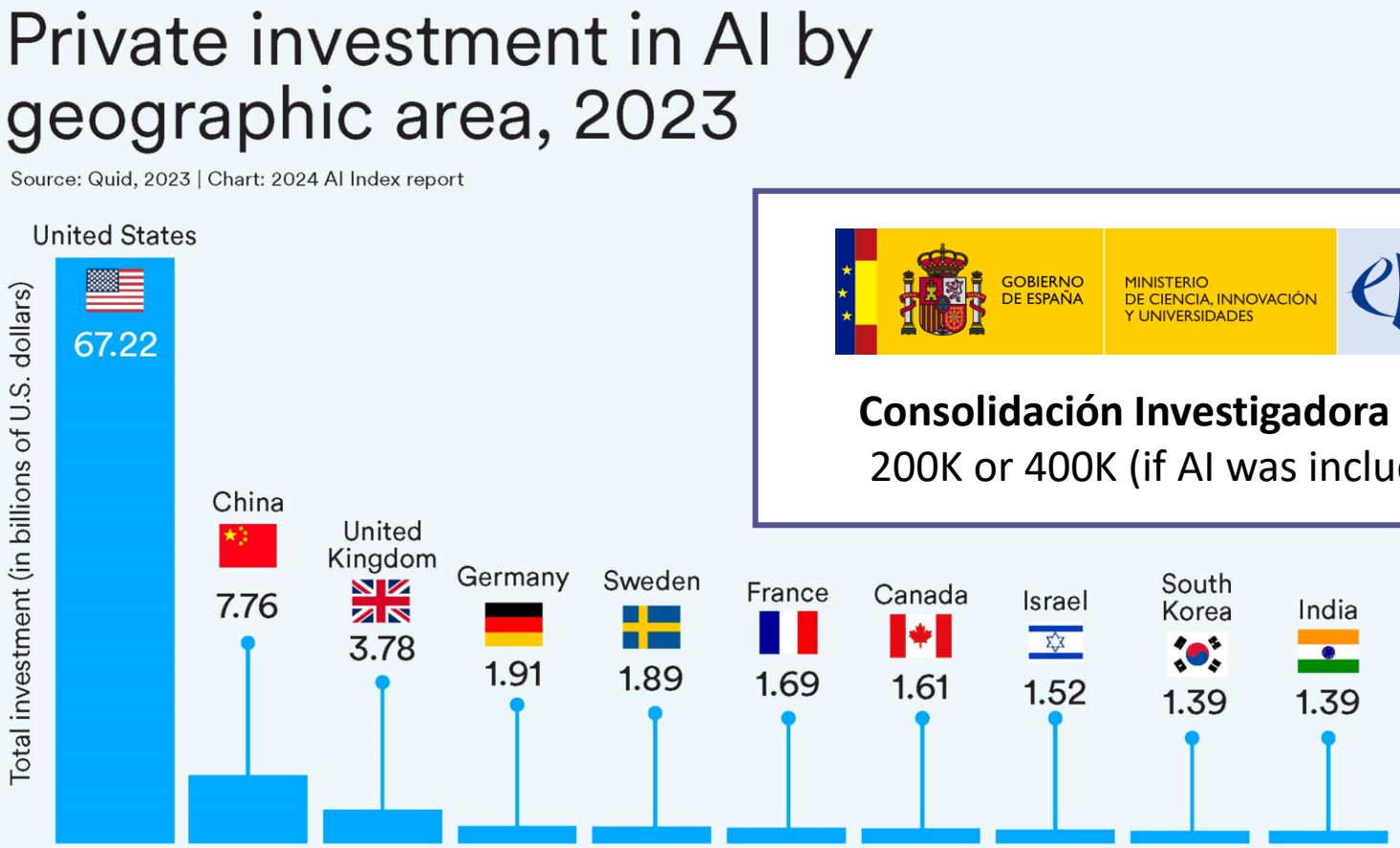

GOBIERNO DE ESPAÑA | MINISTERIO DE CIENCIA, INNOVACIÓN Y UNIVERSIDADES | AGENCIA ESTATAL DE INVESTIGACIÓN

Consolidación Investigadora 2023
200K or 400K (if AI was included)

CSIC
CONSEJO SUPERIOR DE INVESTIGACIONES CIENTÍFICAS

Momentum 2024
2 contracts (AI or digital.)

Spain
0.36

Stanford AI Index Report 2024

Basics of Chemical ML

How does machine learning work?

Data in

ChatGPT

"In a village in la Mancha, whose name I do not care to remember, an hidalgo lived not long ago, one of those who keeps a lance on the rack, an old leather shield, skinny nag..."

"To be, or not to be: that is the question: Whether 'tis nobler in the mind to suffer the slings and arrows of outrageous fortune, or to take arms against a sea of troubles, and by opposing end them? To die: to sleep…

ChatGPT, personal communication, February 11, 2023

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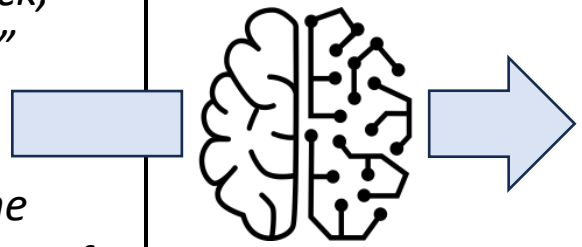

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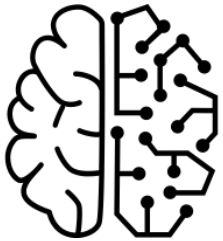
Where does Don Quixote begin?



Who wrote Hamlet?

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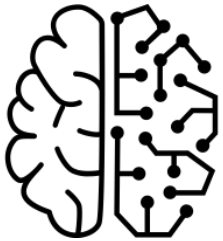

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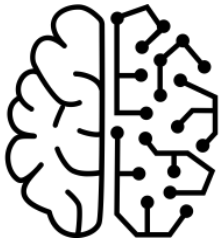

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"William Shakespeare wrote Hamlet. The tragedy of Hamlet, Prince of Denmark (original title in English: The Tragedy of Hamlet, Prince of Denmark), is a play by the…"

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How does machine learning work in chemistry?

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Science **2018**, *360*, 186

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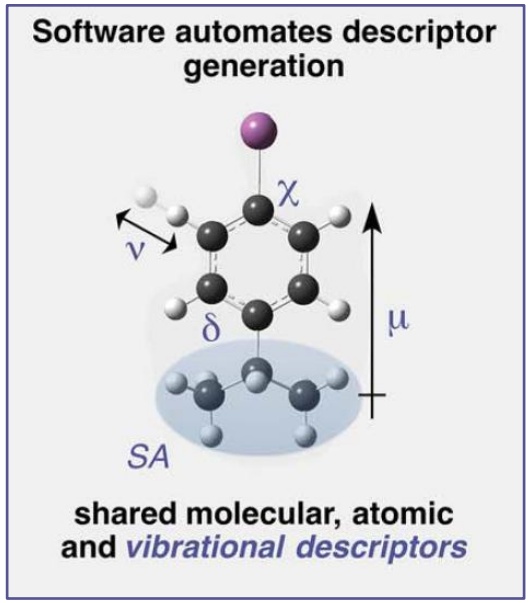

Molecular properties

- *Solubility*
- *m.p.*

Atomic properties

- *Charge*
- *Sterics*

Material properties

- *Component ratio*

Process properties

- *Temperature*
- *Pressure*

Science **2018**, *360*, 186

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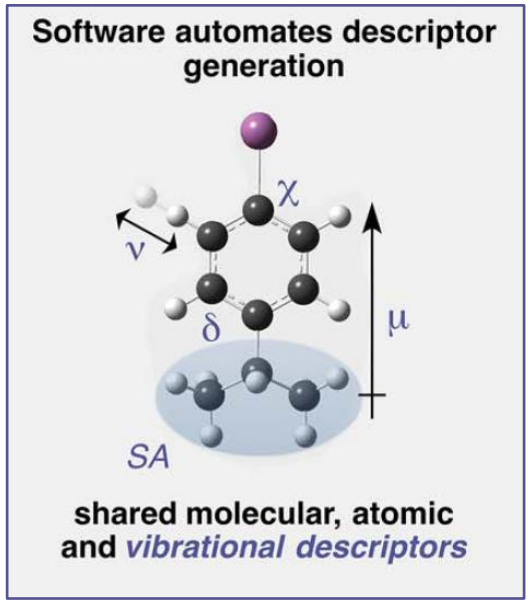

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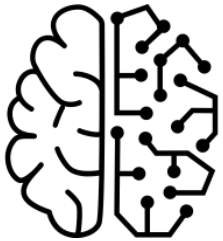

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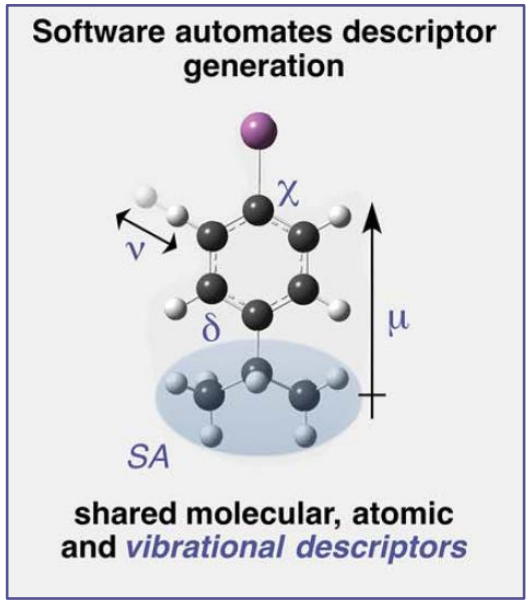

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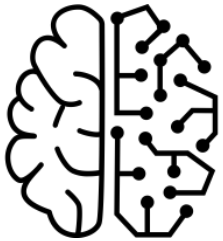

Design an optimal solar cell

Data out

Science **2018**, *360*, 186

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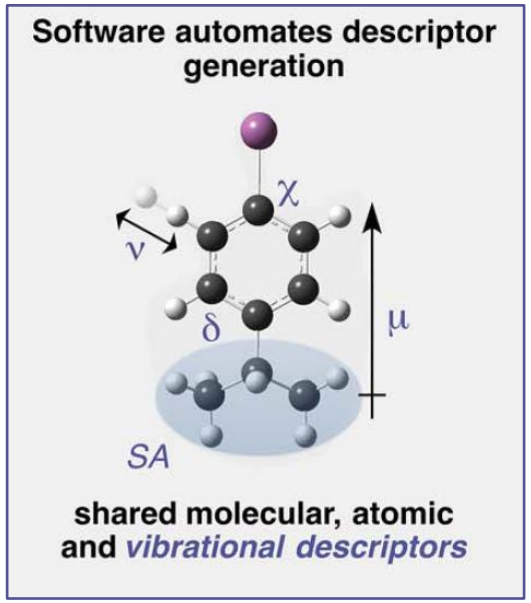

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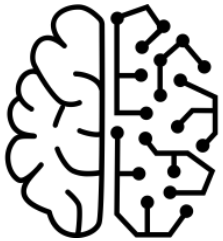

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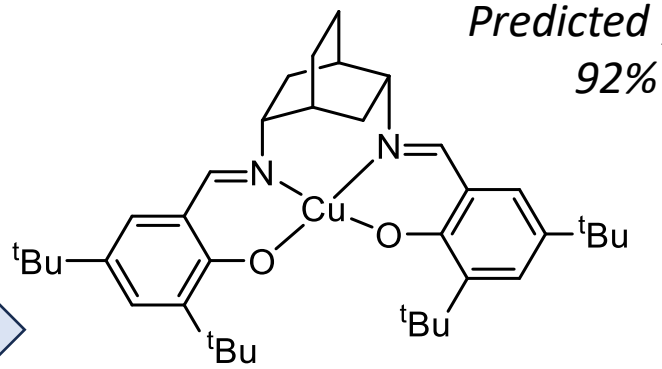

Predicted yield: 92%

Science **2018**, *360*, 186

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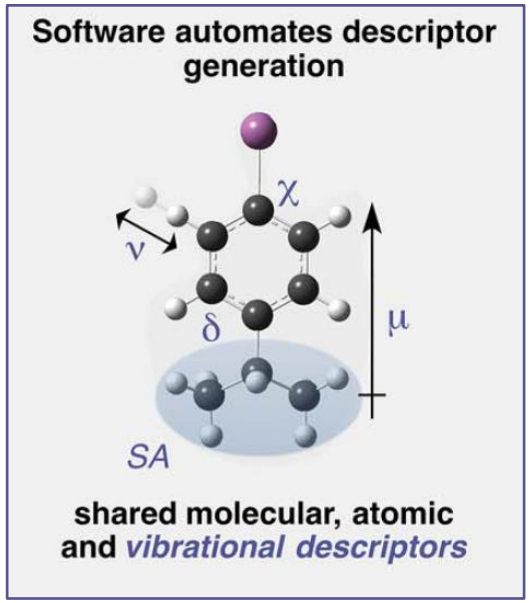

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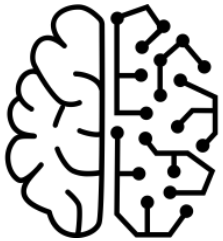

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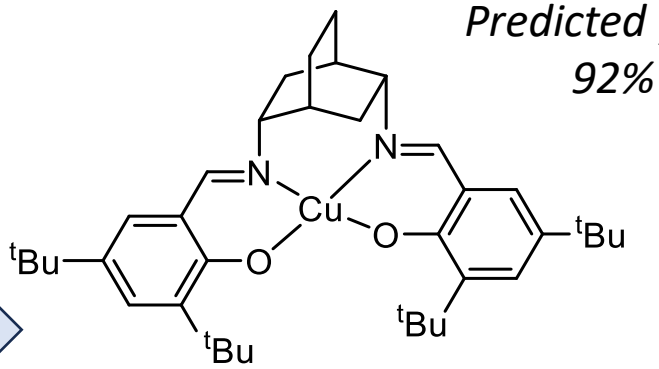


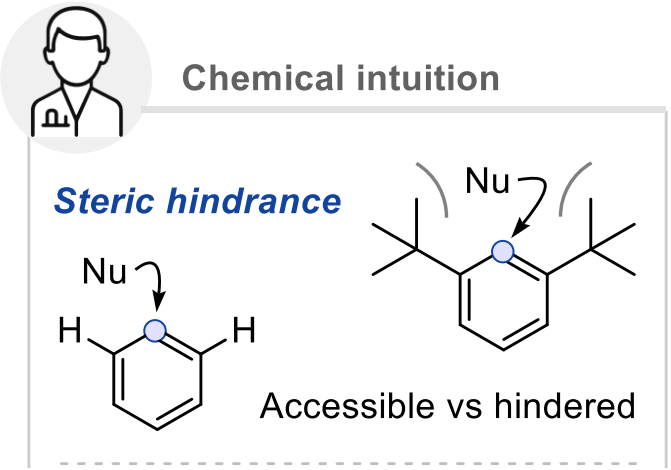
20% graphene + 40% Si + 40% GaAs

Predicted solar-cell efficiency: 21%

Science **2018**, *360*, 186

Basics of Chemical ML

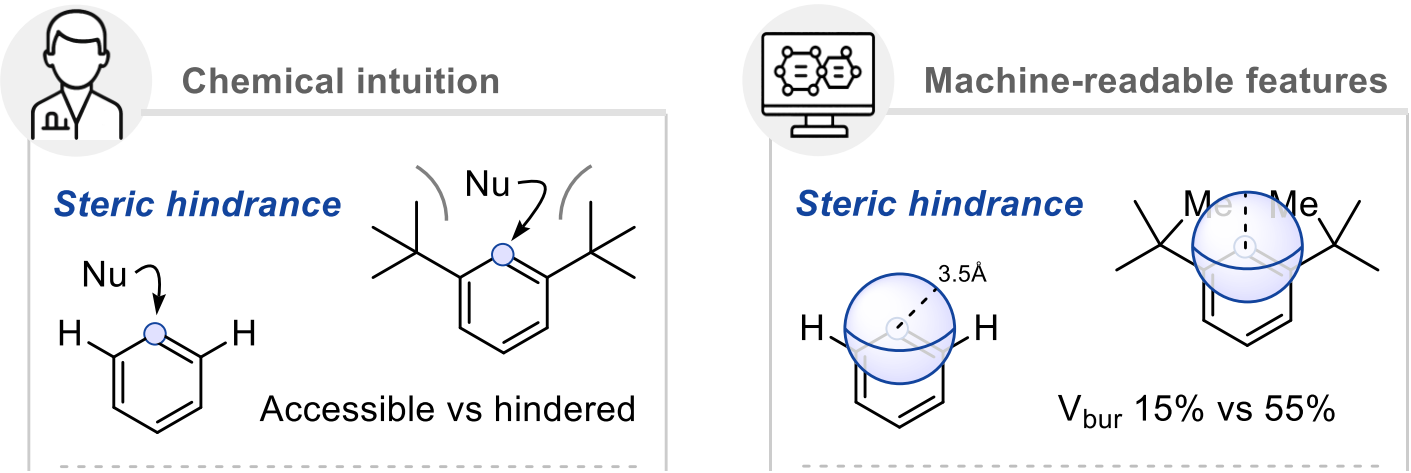
First step: digitalization of molecules



ACS Catal. **2026**, *16*, 1

Basics of Chemical ML

First step: digitalization of molecules



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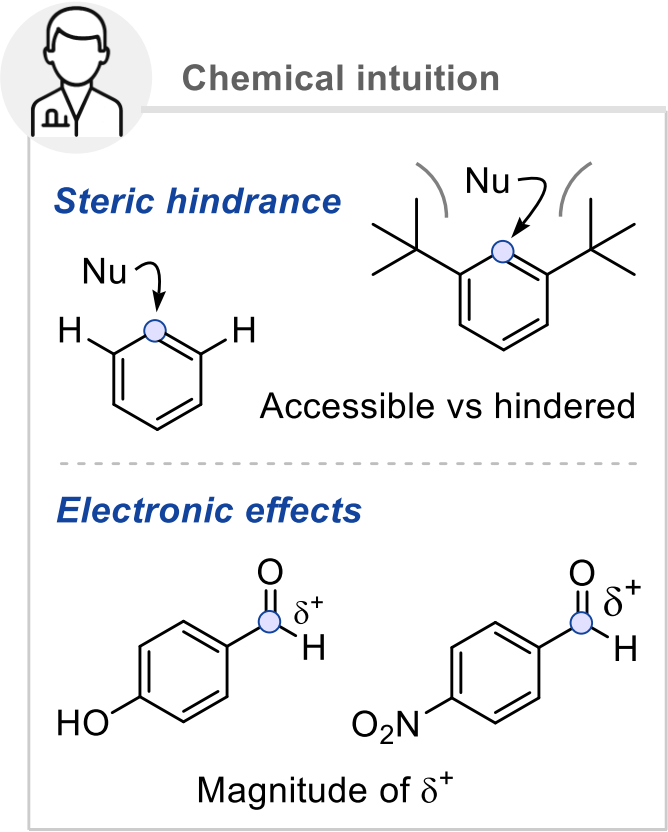

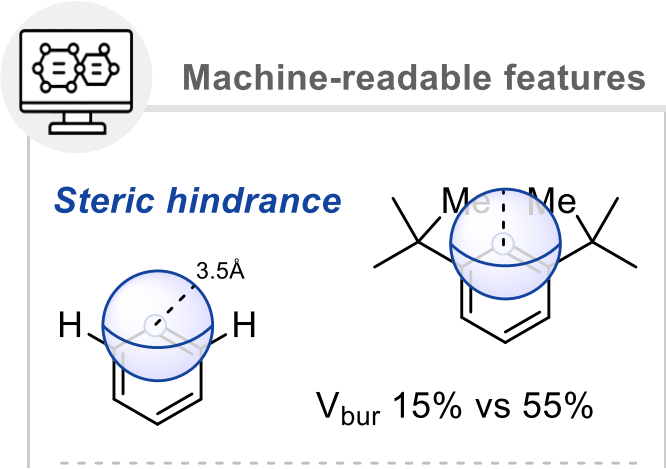

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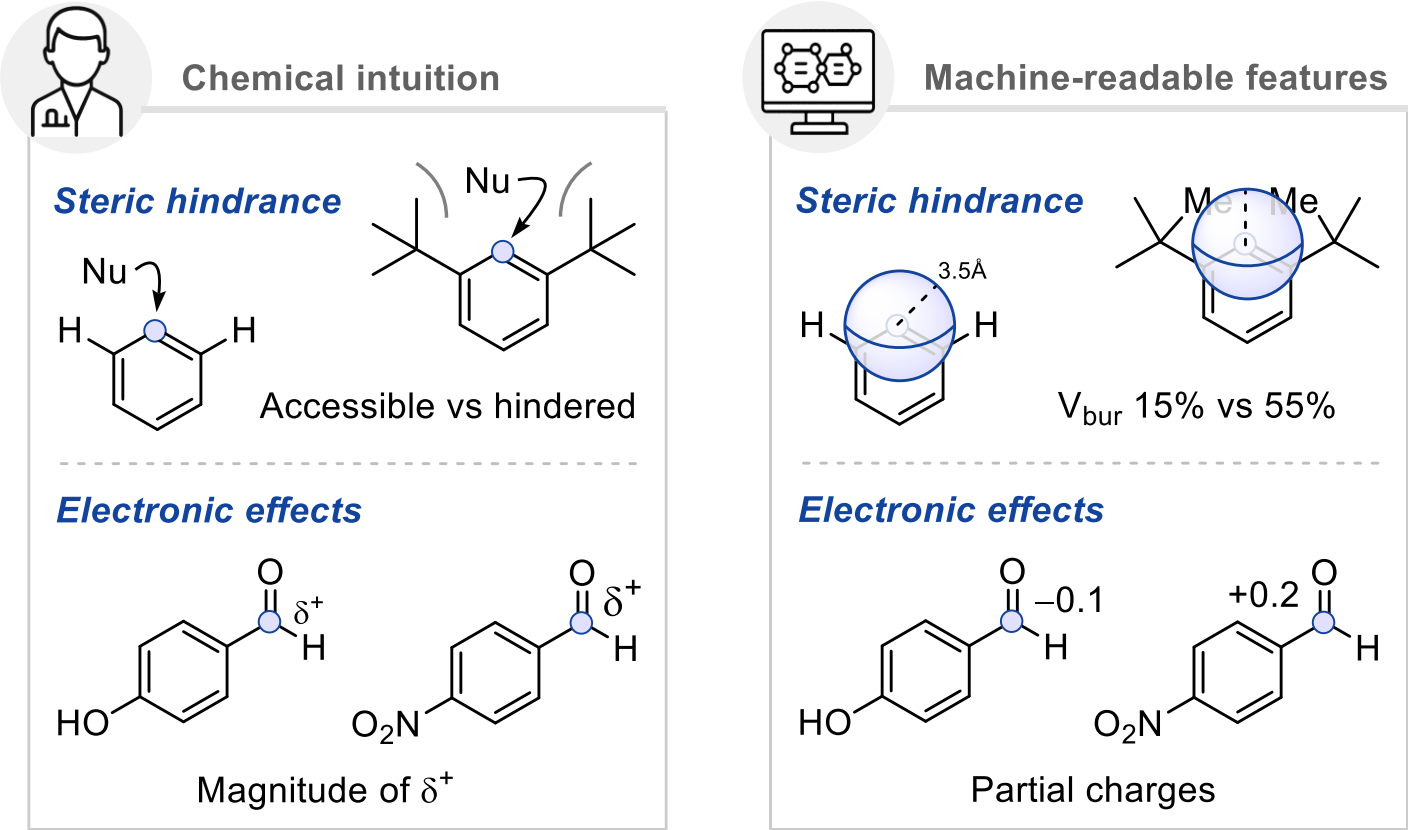


ACS Catal. **2026**, *16*, 1

Basics of Chemical ML

First step: digitalization of molecules

Guidelines
Generate or collect relevant atomic and molecular descriptors to represent your system. A few options: • Databases: KRAKEN,[65] ioChem-BD,[66] OSCAR,[67] and OMol25.[68] • Topological/MM descriptors and fingerprints: RDKit,[17] CDK,[69] and PaDELPy.[70] • QM descriptors: AQME,[71] ObeLiX,[72] Autoqchem,[73] and molli.[74] • Steric descriptors: MORFEUS,[75] DBSTEP,[76] and SambVca.[77] • ML-predicted properties: Chemprop[78] and MolPROP.[79]

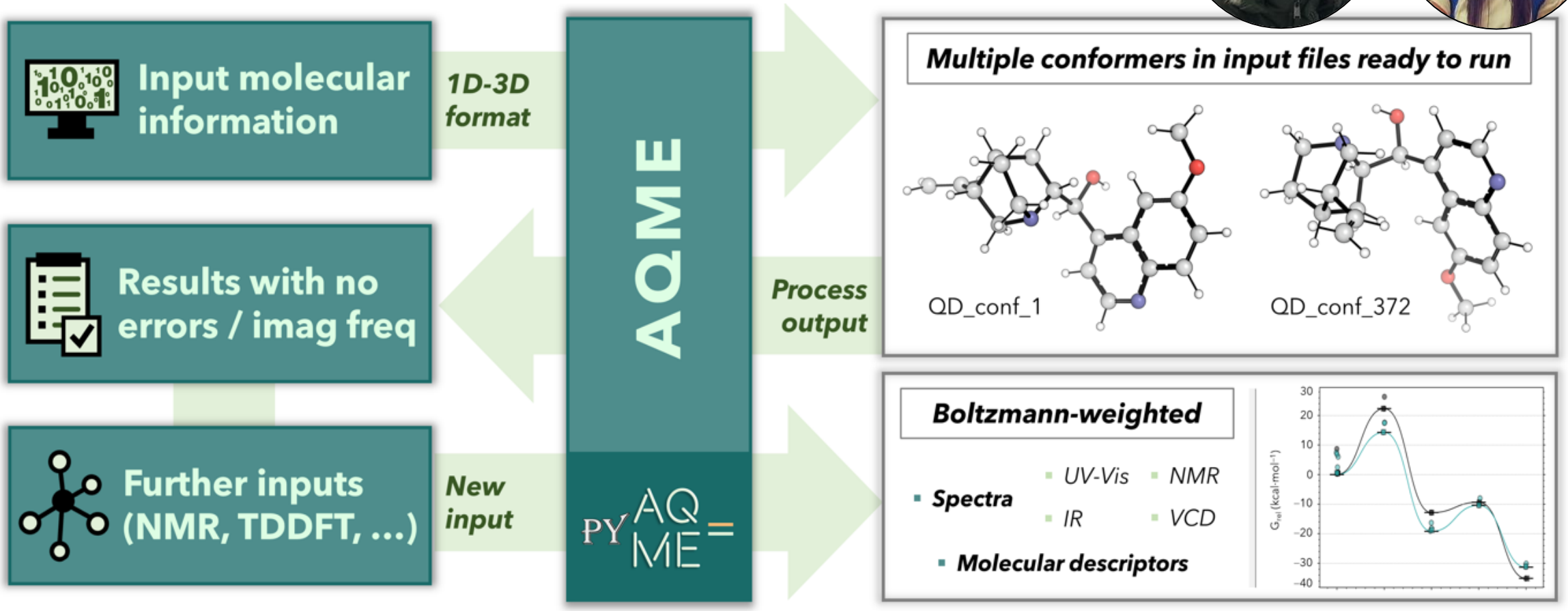
ACS Catal. **2026**, *16*, 1

AQME: automation of QM protocols

Overview of AQME

Shree Sowndarya

Dr. Robert S. Paton

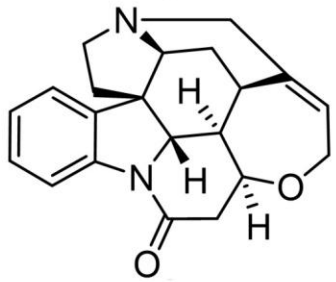


WIREs. Comput. Mol. Sci. **2023**, e1663

AQME: automation of QM protocols

End-to-end workflows

Input: SMILES string



RDKit — CSEARCH → Gaussian — QPREP → cclib — QCORR → Gaussian — QDESCP

Output:

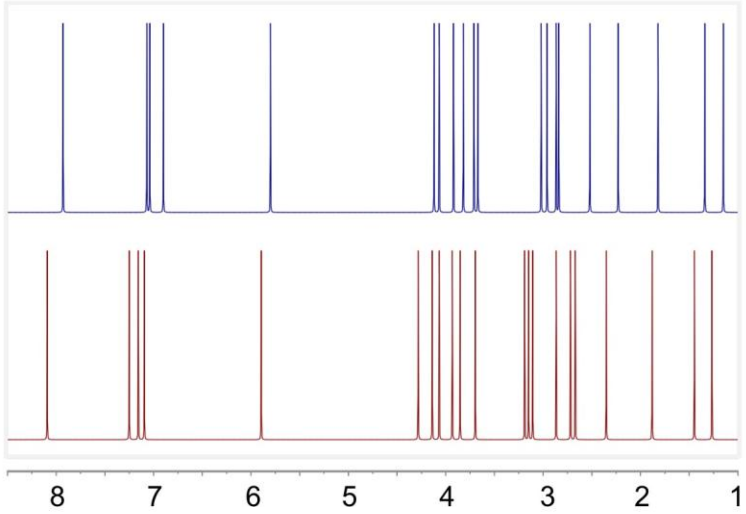
- Boltzmann weighted NMR spectra
- Boltzmann conformer distribution

Automated tasks in this workflow

- Conformational search with RDKit
- Gaussian input generation & submission
- Error fixing of QM outputs
- GoodVibes conformer ratio calculation
- Estimation of NMR shifts in TMS scale

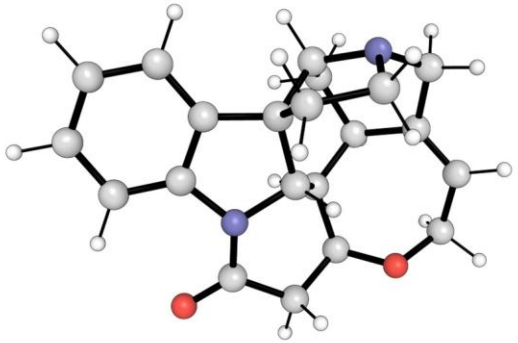
^{1}H NMR spectra

Experim. vs calcul. MAE = 0.14 ppm



Conformer ratio

Experim. 97:3 vs calcul. 99:1



C1CN2CC3=CCO[C@H]4CC(=O)N5[C@H]6[C@H]4[C@H]3C[C@H]2[C@@]61C7=CC=CC=C75

sbatch AQME_script.sh

AQME_script.sh

Alegre-Requena, Paton & col., *WIREs. Comput. Mol. Sci.* **2023**, e1663

AQME: stats

Users and interest

https://github.com/jvalegre/aqme

www.readthedocs.org/projects/aqme

Downloads 39k

Known active users:

Academic groups

Dr. A. Vazquez-Mayagoitia
Argonne National Lab.

Prof. Juho Helaja
University of Helsinki

C-CAS

YouTube ES

The Alegre Group

@thealegregroup4964 · 97 subscribers · 7 videos

Companies

gsk abbvie Lilly syngenta

WIREs. Comput. Mol. Sci. **2023**, e1663

AQME: automation of QM protocols

iSQCH
Instituto de Síntesis Química y Catálisis Homogénea

THE Alegre GROUP

A. Conformational search

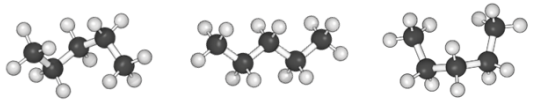
Manual approach

- Open 3D visualization tool
- Draw conformers
- All relevant conformers?

AQME (CSEARCH)

- Execute command line:

python -m aqme --csearch --program rdkit --smi CCCCC --name pentane



B. Input file creation

Manual approach

- Insert keywords
- Input coordinates
- Add extra lines

AQME (QPREP)

- Execute command line:

python -m aqme --qprep --files “*.log” --qm_input "wB97XD/6/31+G(d)" --program gaussian

C	0.1223	1.2342	0.0000
C	0.2456	0.8201	1.83...

C. Post-processing of QM outputs

Manual approach

- Open QM output files
- Check termination status
- Fix termination errors

AQME (QCORR)

- Execute command line:

python -m aqme --qcorr --files "*.log"

- Normal
- Opt. conv.
- Imag. freqs
- ...

D. Generation of molecular descriptors

Manual approach

- Run calculations
- Retrieve properties
- Compile database

AQME (QDESCP)

- Execute command line:

python -m aqme --qdescp --program xtb --files "*.sdf"

- Dipole
- Charges
- FOD
- Homo/LUMO
- ...

Future Leaders Special Issue. *Trends in Chem.* **2024**

AQME: automation of QM protocols

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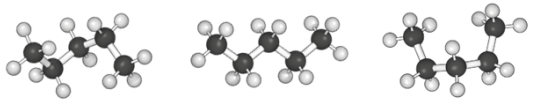

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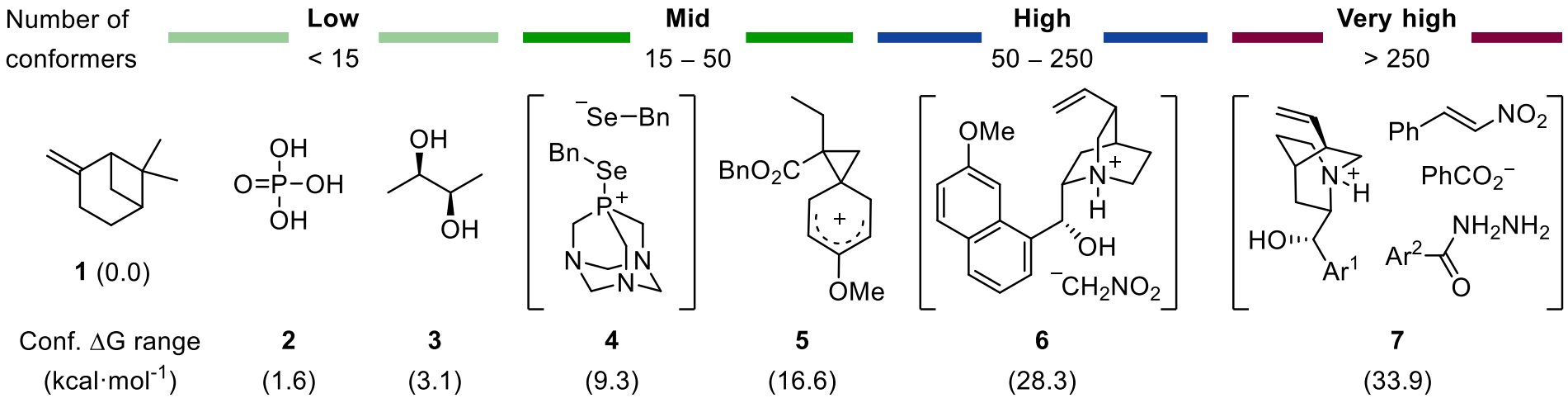
Future Leaders Special Issue. *Trends in Chem.* **2024**

AQME: automation of QM protocols

Conformer searches are tricky



What's the most stable conformer in each case?

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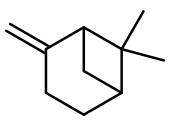
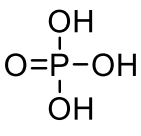
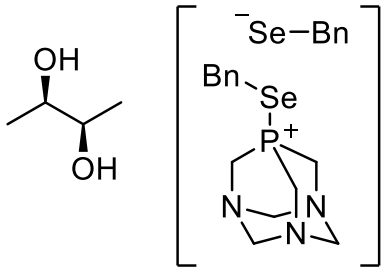
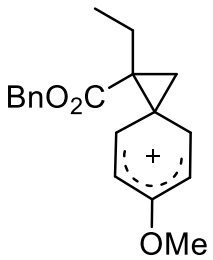
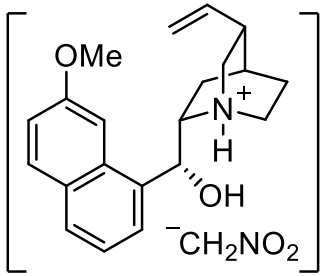
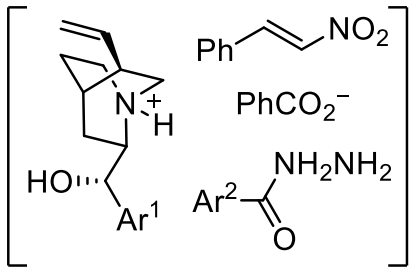
4: *Nature Commun.* **2022**, *13*, 6885.
5: *J. Am. Chem. Soc.* **2022**, *144*, 86.
6: *Chem. Eur. J.* **2017**, *23*, 15336
7: *ChemRxiv*, **2023**, 10.26434/chemrxiv-2023-8jzsp

AQME: automation of QM protocols

Conformer searches are tricky

Number of conformers	Low < 15		Mid 15 – 50		High 50 – 250	Very high > 250
						
	1 (0.0)	**2**	**3**	**4**	**5**	**6**
Conf. ΔG range (kcal·mol^{-1})	(1.6)	(3.1)	(9.3)	(16.6)	(28.3)	**7** (33.9)

Conformational sampling strategy

Manual	✓	~	X	X
Automated	✓	✓	~	X
Multi-seeding auto.	✓	✓	✓	~

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AQME: automation of QM protocols

iSQCH
Instituto de Síntesis Química y Catálisis Homogénea

THE Alegre GROUP

A. Conformational search

Manual approach

· Open 3D visualization tool

· Draw conformers

· All relevant conformers?

AQME (CSEARCH)

· Execute command line:

python -m aqme --csearch --program rdkit --smi CCCCC --name pentane

B. Input file creation

Manual approach

· Insert keywords

· Input coordinates

· Add extra lines

AQME (QPREP)

· Execute command line:

python -m aqme --qprep --files “*.log” --qm_input "wB97XD/6/31+G(d)" --program gaussian

C	0.1223	1.2342	0.0000
C	0.2456	0.8201	1.83...

C. Post-processing of QM outputs

Manual approach

· Open QM output files

· Check termination status

· Fix termination errors

AQME (QCORR)

· Execute command line:

python -m aqme --qcorr --files "*.log"

· Normal · Opt. conv.
· Imag. freqs · ...

D. Generation of molecular descriptors

Manual approach

· Run calculations

· Retrieve properties

· Compile database

AQME (QDESCP)

· Execute command line:

python -m aqme --qdescp --program xtb --files "*.sdf"

· Dipole · Charges · FOD
· Homo/LUMO · ...

Dr. Kjell Jorner

MORFEUS

xTB

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